Homework 4 – Regression Consult

When people make decisions, their choices are often swayed by how information is presented. This is often described as "the framing effect." One example of this, is how commercials describe new medications. As an example, Nereus (tradipitant), a new drug for motion-induced vomiting like sea-sickness, is marketed with a **positive frame** as reducing vomiting incidence from 44% to 18% (a reduction of 60%). This marketing makes it seem more effective than a **negative frame** that says just 18% of patients still experienced vomiting, even though both statements are true.

Xu (2024) aimed to evaluate whether older adults are more or less vulnerable to framing than younger adults. They collected a sample of 696 adults aged 18 to 71 and exposed them to three different scenarios

Scenario 1: Rate Drug Efficacy (1 = very bad, 6 = very good)

- **Positive Frame:** 70 out of 100 patients felt better
- **Negative Frame:** 30 out of 100 patients did not feel better

Scenario 2: Likelihood to stop drinking cocoa (1 = surely stop, 6 = surely continue)

- **Positive Frame:** Stopping drinking hot cocoa reduces cadmium and cancer risk.
- **Negative Frame:** Continuing to drink hot cocoa increases cadmium and cancer risk.

Scenario 3: Preference for chance choice (1 = surely safe option, 6 = surely risky)

- **Positive Frame:** Choice between 200 saved vs 1/3 chance all 600 are saved.
- **Negative Frame:** Choice between 400 die vs 2/3 chance all 600 die.

In each scenario, participants were randomly assigned a positive or negative frame. They rated their response on a 1-6 scale. The researchers want to build three models, one for each scenario adjusting for simultaneous linear changes in demographic variables (gender, ethnicity, education level, political affiliation).

They want to evaluate whether, as adults age, the way choices are worded have more influence on their judgments and ultimate choices. If there is a change, are those changes driven by positive or negative framings? At what age does this effect onset?

Your Goals:

- Determine the correct analysis strategy and conduct the necessary analyses.
  - Hint 1: Ensure to assess and communicate results about the model fit.
  - Hint 2: Ensure to think about whether or which interactions need to be included.
  - Hint 3: You will need run various post hoc procedures (e.g., `emmeans`, `emtrends` or `johnson_neyman`).
  - Hint 4: Looking at the paper may be helpful. They do include `R` code, but I expect you to do a much better job using the techniques we discussed (they do somethings differently)
- Write a clear, concise, complete, and correct report conveying an answer to their research question.

The data are in `HW4Data.csv` but they will require some manual and `tidyverse` cleaning.

1 Loading and Cleaning the Data

```
dat.raw <- read_csv("HW4Data.csv", name_repair = "unique_quiet")
new.names <- unlist(dat.raw[1, ], use.names = FALSE)

dat <- dat.raw[-1, !is.na(new.names)]
names(dat) <- str_trim(new.names[!is.na(new.names)])
dat <- dat |>
```

```
  type_convert()

dim(dat)
```

```
[1] 842  32
```

```
dat <- dat |>
  filter(finished == TRUE)

nrow(dat)
```

```
[1] 696
```

```
table(dat$attentioncheck, useNA = "always")
```

```
   1    4    6 <NA>
 650    2    1   43
```

```
dat <- dat |>
  mutate(agecheck = stringi::stri_trans_nfkc(ageattentioncheck),
         agecheck.lo = as.numeric(str_sub(agecheck, 1, 2)),
         agecheck.hi = as.numeric(str_sub(agecheck, 4, 5)),
         age.ok = age >= agecheck.lo & age <= agecheck.hi)

table(dat$age.ok, useNA = "always")
```

```
FALSE  TRUE  <NA>
    5   596    95
```

```
dat <- dat |>
  filter(attentioncheck == 1 & age.ok == TRUE)

nrow(dat)
```

```
[1] 592
```

```
dat <- dat |>
  mutate(frame1 = factor(ifelse(treatment1 == "pos1", "pos", "neg"), levels = c("pos", "neg")),
         frame2 = factor(ifelse(treatment2 == "pos2", "pos", "neg"), levels = c("pos", "neg")),
         frame3 = factor(ifelse(treatment3 == "pos3", "pos", "neg"), levels = c("pos", "neg")))
```

```
table(dat$gender)
```

```
                  Female                     Male Non-binary / third gender
                     349                      233                        7
       Prefer not to say
                       3
```

```
table(dat$ethnicity)
```

```
   American Indian or Alaska Native                                 Asian
                                  2                                    43
          Black or African American Native Hawaiian or Pacific Islander
                                 46                                     2
                           Other                   Prefer not to say
                                16                                     3
                           White
                             480
```

```r
table(dat$education)
```

```
       2 year degree         4 year degree             Doctorate
                  65                   233                     9
High school graduate Less than high school   Professional degree
                  59                     2                   113
        Some college
                 111
```

```r
table(dat$politics)
```

```
        Democrat       Independent Prefer not to say        Republican
             256               168                 6               140
   Something else
              22
```

```r
dat <- dat |>
  mutate(gender = case_when(gender %in% c("Female", "Male") ~ gender,
                            TRUE ~ "Other"),
         ethnicity = case_when(ethnicity %in% c("White", "Black or African American", "Asian") ~ ethnicity,
                               TRUE ~ "Other"),
         education = case_when(education %in% c("Less than high school",
                                                "High school graduate") ~ "High school or less",
                               education %in% c("Professional degree",
                                                "Doctorate") ~ "Graduate degree",
                               TRUE ~ education),
         politics = case_when(politics %in% c("Democrat", "Republican", "Independent") ~ politics,
                              TRUE ~ "Other")) |>
  mutate(gender = factor(gender, levels = c("Female", "Male", "Other")),
         ethnicity = factor(ethnicity, levels = c("White", "Black or African American",
                                                  "Asian", "Other")),
         education = factor(education, levels = c("High school or less", "Some college",
                                                  "2 year degree", "4 year degree",
                                                  "Graduate degree")),
         politics = factor(politics, levels = c("Democrat", "Republican",
                                                "Independent", "Other")))

table(dat$gender)
```

```
Female   Male  Other
   349    233     10
```

```r
table(dat$ethnicity)
```

```
                   White Black or African American                   Asian
                     480                       46                      43
                   Other
                      23
```

```r
table(dat$education)
```

```
High school or less        Some college       2 year degree       4 year degree
                 61                 111                  65                 233
    Graduate degree
                122
```

```r
table(dat$politics)
```

```
   Democrat  Republican Independent       Other
        256         140         168          28
```

```r
dat <- dat |>
  select(pick1, pick2, pick3, frame1, frame2, frame3,
         age, gender, ethnicity, education, politics)

summary(dat)
```

```
     pick1           pick2           pick3       frame1    frame2    frame3
 Min.   :2.00   Min.   :1.000   Min.   :1.000   pos:274   pos:295   pos:303
 1st Qu.:4.00   1st Qu.:1.000   1st Qu.:2.000   neg:318   neg:297   neg:289
 Median :4.00   Median :2.000   Median :3.000
 Mean   :4.12   Mean   :2.262   Mean   :3.215
 3rd Qu.:5.00   3rd Qu.:3.000   3rd Qu.:4.000
 Max.   :6.00   Max.   :6.000   Max.   :6.000
                                NA's   :1
      age          gender                          ethnicity
 Min.   :18.0   Female:349   White                      :480
 1st Qu.:31.0   Male  :233   Black or African American: 46
 Median :38.0   Other : 10   Asian                      : 43
 Mean   :40.8                Other                      : 23
 3rd Qu.:50.0
 Max.   :71.0

               education          politics
 High school or less: 61   Democrat   :256
 Some college       :111   Republican :140
 2 year degree      : 65   Independent:168
 4 year degree      :233   Other      : 28
 Graduate degree    :122
```

The raw export had two header rows and 20 empty trailing columns; I kept the variable-name row as headers and dropped the empty columns. Of 842 rows, 696 were completed surveys. I kept only the 592 participants who passed both attention checks. I set the positive frame as the reference level, so the interaction term estimates how the age effect differs under the negative frame. I collapsed sparse demographic categories: gender (Female, Male, Other), ethnicity (White, Black, Asian, Other), education (High school or less, Some college, 2 year, 4 year, Graduate), and politics (Democrat, Republican, Independent, Other).

2 Exploratory Data Analyses

```r
dat |>
  group_by(frame1) |>
  summarize(n = n(), Mean = mean(pick1), SD = sd(pick1)) |>
  knitr::kable(digits = 3)
```

frame1	n	Mean	SD
pos	274	4.467	0.630
neg	318	3.821	0.815

```r
dat |>
  group_by(frame2) |>
  summarize(n = n(), Mean = mean(pick2), SD = sd(pick2)) |>
  knitr::kable(digits = 3)
```

frame2	n	Mean	SD
pos	295	2.549	1.423
neg	297	1.976	1.060

```r
dat |>
  group_by(frame3) |>
  summarize(n = n(), Mean = mean(pick3, na.rm = TRUE), SD = sd(pick3, na.rm = TRUE)) |>
  knitr::kable(digits = 3)
```

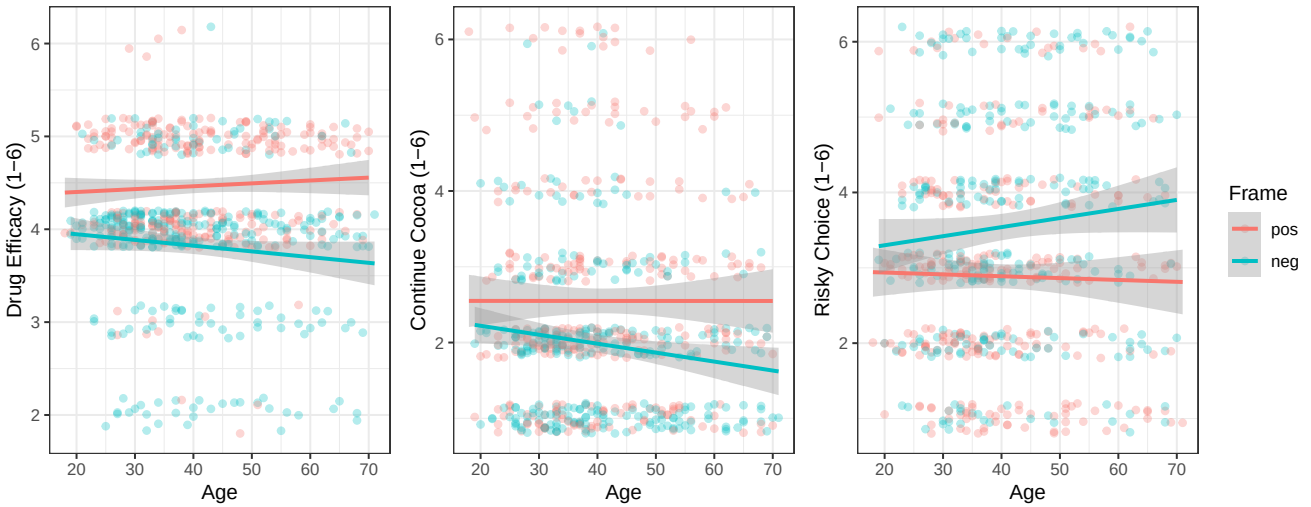
frame3	n	Mean	SD
pos	303	2.888	1.407
neg	289	3.559	1.508

```r
p1 <- ggplot(dat, aes(x = age, y = pick1, color = frame1)) +
  geom_jitter(width = 0, height = 0.2, alpha = 0.3) +
  geom_smooth(method = "lm") +
  theme_bw() +
  xlab("Age") +
  ylab("Drug Efficacy (1-6)") +
  labs(color = "Frame")

p2 <- ggplot(dat, aes(x = age, y = pick2, color = frame2)) +
  geom_jitter(width = 0, height = 0.2, alpha = 0.3) +
  geom_smooth(method = "lm") +
  theme_bw() +
  xlab("Age") +
  ylab("Continue Cocoa (1-6)") +
  labs(color = "Frame")

p3 <- ggplot(dat, aes(x = age, y = pick3, color = frame3)) +
  geom_jitter(width = 0, height = 0.2, alpha = 0.3) +
  geom_smooth(method = "lm") +
  theme_bw() +
  xlab("Age") +
  ylab("Risky Choice (1-6)") +
  labs(color = "Frame")

(p1 + p2 + p3) + plot_layout(guides = "collect")
```




```r
t.test(age ~ frame1, data = dat)
```

```
    Welch Two Sample t-test

data:  age by frame1
t = 0.81247, df = 581.19, p-value = 0.4169
alternative hypothesis: true difference in means between group pos and group neg is not equal to 0
95 percent confidence interval:
 -1.181506  2.848678
sample estimates:
mean in group pos mean in group neg
         41.25182          40.41824
```

```r
t.test(age ~ frame2, data = dat)
```

```
    Welch Two Sample t-test

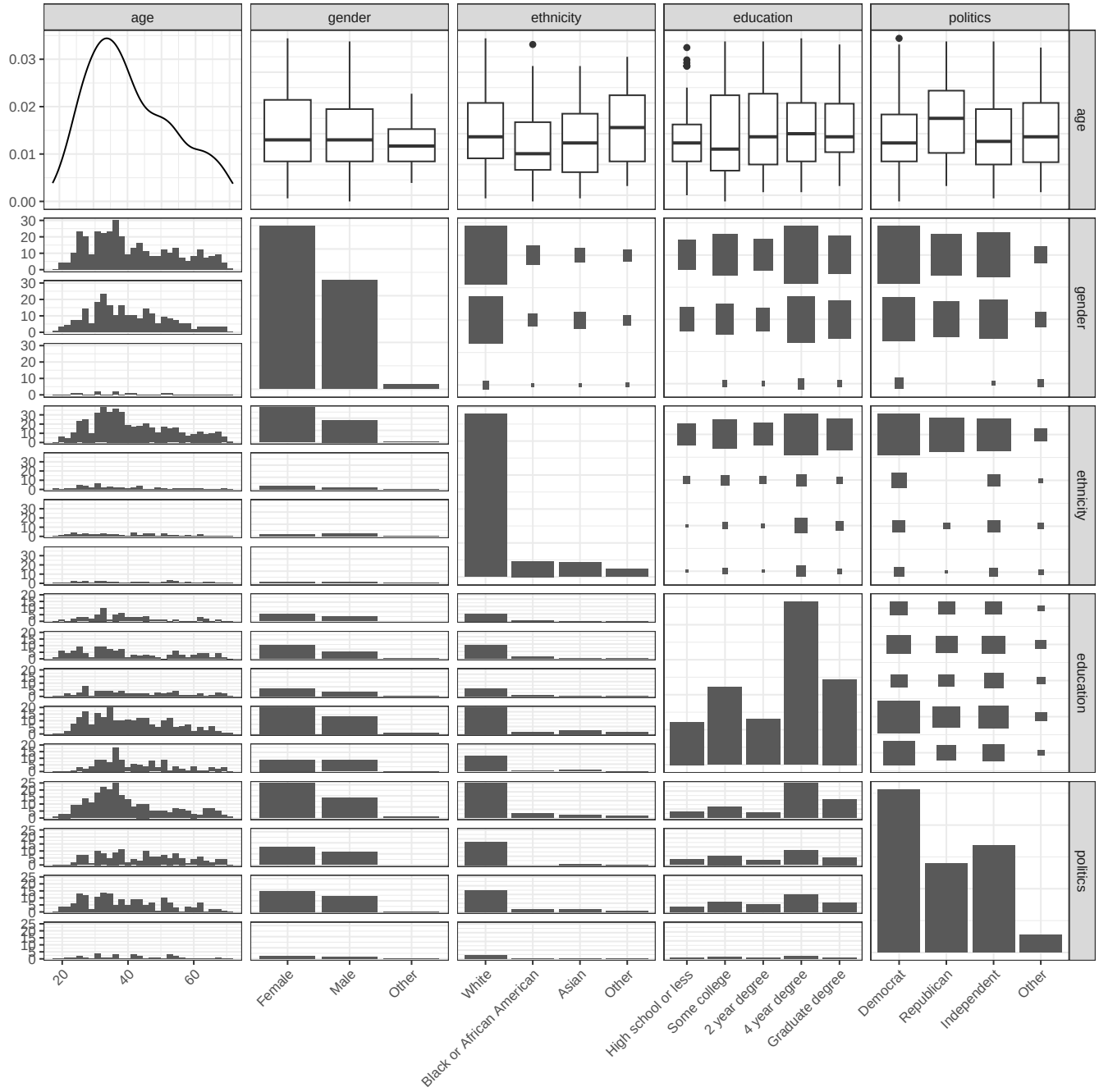
data:  age by frame2
t = 0.16341, df = 589.85, p-value = 0.8703
alternative hypothesis: true difference in means between group pos and group neg is not equal to 0
95 percent confidence interval:
 -1.846719  2.181913
sample estimates:
mean in group pos mean in group neg
         40.88814          40.72054
```

```r
t.test(age ~ frame3, data = dat)
```

```
    Welch Two Sample t-test

data:  age by frame3
t = -1.5555, df = 588.13, p-value = 0.1204
alternative hypothesis: true difference in means between group pos and group neg is not equal to 0
95 percent confidence interval:
 -3.6043605  0.4184115
sample estimates:
mean in group pos mean in group neg
         40.02640          41.61938
```

```r
GGally::ggpairs(dat, columns = c("age", "gender", "ethnicity", "education", "politics")) +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



I first compared mean responses by frame. The drug was rated as more effective under the positive frame, participants were less willing to continue drinking cocoa under the negative frame, and they preferred the risky option more under the negative frame. Plotting responses against age by frame showed that the positive-frame lines were nearly flat, while the negative-frame lines changed with age, so the gap between frames appeared to widen for older adults; this suggests a frame × age interaction driven by the negative frame. Welch t-tests found no age difference between frames in any scenario, confirming that random assignment balanced age. Pairwise plots showed no clear associations among the predictors, so multicollinearity does not appear to be a concern.

3 Regression Analyses

3.1 Rating Drug Efficacy

```
source("plotResiduals.R")
source("plotInfluence.R")
```

```
mean.age <- mean(dat$age)

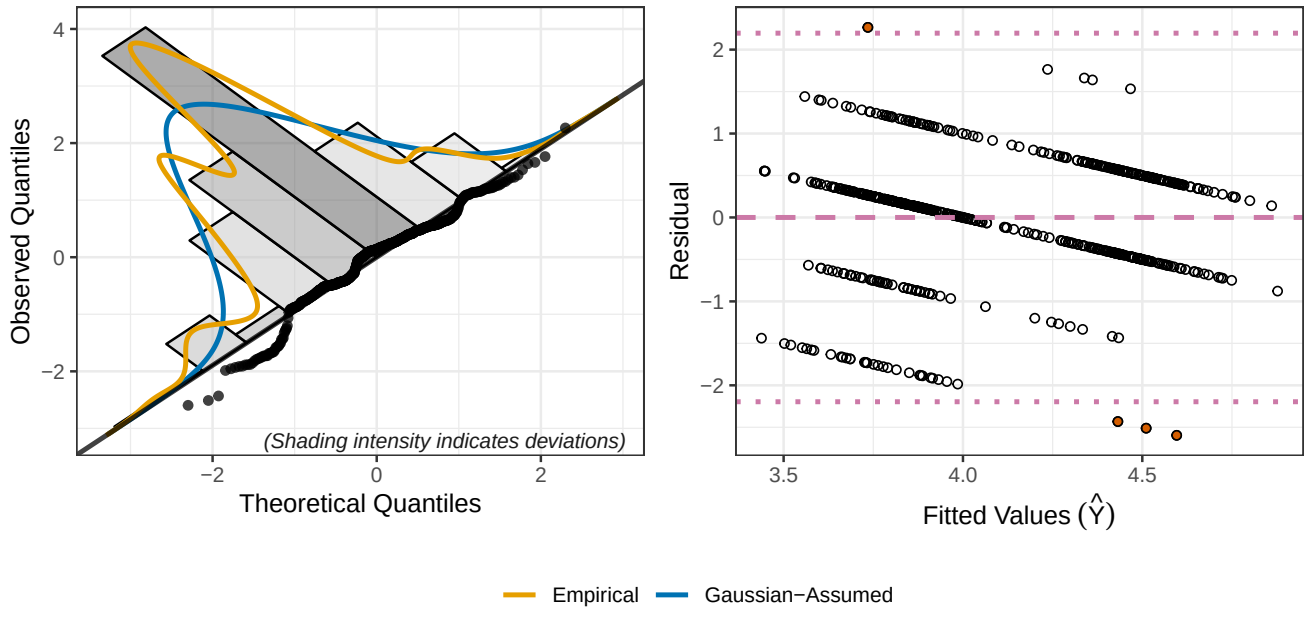
dat <- dat |>
  mutate(age.c = age - mean.age)

mean.age
```

```
[1] 40.80405
```

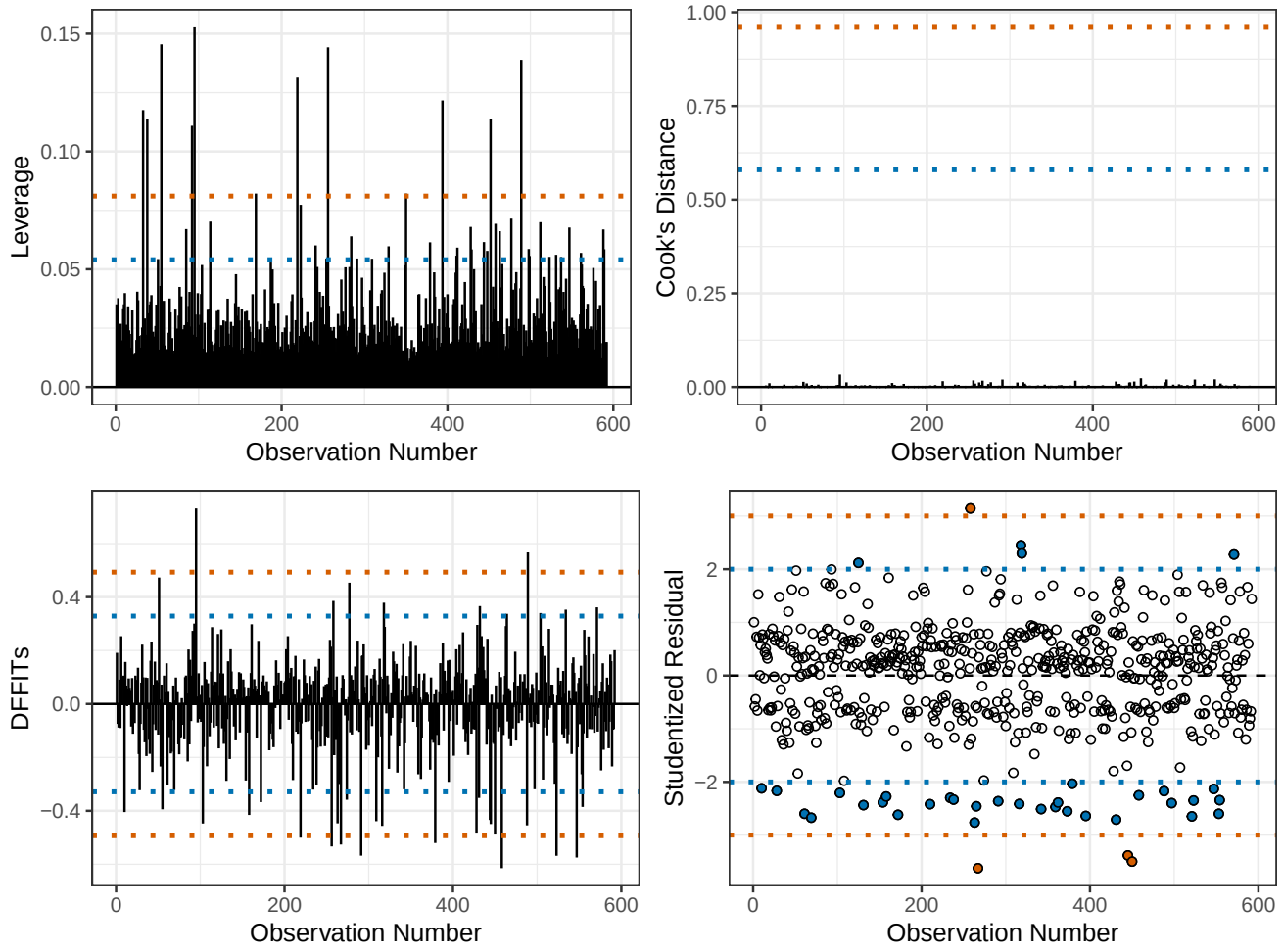
```
mod1 <- lm(pick1 ~ frame1 * age.c + gender + ethnicity + education + politics,
           data = dat)
```

```
plotResiduals(mod1)
```



The residuals fall along parallel bands because responses can only be integers from 1 to 6. The Q-Q plot shows that the residuals are close to Gaussian.

```
plotInfluence(mod1)
```



Several observations have high leverage, likely because they belong to small demographic groups. However, all Cook's distances are far below the moderate threshold, so no single observation has a strong influence on the model.

```
car::vif(mod1)
```

```
                GVIF Df GVIF^(1/(2*Df))
frame1      1.017894  1        1.008907
age.c       2.339811  1        1.529644
gender      1.075843  2        1.018444
ethnicity   1.143267  3        1.022566
education   1.133688  4        1.015808
politics    1.185533  3        1.028772
frame1:age.c 2.264757  1        1.504911
```

```
car::vif(lm(pick1 ~ frame1 + age.c + gender + ethnicity + education + politics,
            data = dat))
```

```
              GVIF Df GVIF^(1/(2*Df))
frame1    1.017889  1        1.008905
age.c     1.055095  1        1.027178
gender    1.073616  2        1.017917
ethnicity 1.137853  3        1.021757
education 1.130613  4        1.015463
politics  1.178447  3        1.027744
```

All adjusted GVIF values were between 1.0 and 1.5, so multicollinearity is not a concern.

```r
anova(mod1)
```

```
Analysis of Variance Table

Response: pick1
              Df  Sum Sq Mean Sq  F value  Pr(>F)
frame1         1  61.497  61.497 114.7912 < 2e-16 ***
age.c          1   0.367   0.367   0.6841 0.40851
gender         2   0.735   0.367   0.6858 0.50410
ethnicity      3   3.262   1.087   2.0298 0.10860
education      4   2.095   0.524   0.9778 0.41906
politics       3   1.714   0.571   1.0666 0.36273
frame1:age.c   1   2.232   2.232   4.1668 0.04168 *
Residuals    576 308.582   0.536
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
sum.mod1 <- summary(mod1)
effectsize::F_to_omega2(f = sum.mod1$fstatistic[1],
                        df = sum.mod1$fstatistic[2],
                        df_error = sum.mod1$fstatistic[3])
```

```
Omega2 (partial) |       95% CI
--------------------------------
0.17             | [0.11, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

```r
effectsize::omega_squared(anova(mod1))
```

```
# Effect Size for ANOVA (Type I)

Parameter    | Omega2 (partial) |       95% CI
-----------------------------------------------
frame1       |             0.16 | [0.12, 1.00]
age.c        |             0.00 | [0.00, 1.00]
gender       |             0.00 | [0.00, 1.00]
ethnicity    |         5.19e-03 | [0.00, 1.00]
education    |             0.00 | [0.00, 1.00]
politics     |         3.37e-04 | [0.00, 1.00]
frame1:age.c |         5.32e-03 | [0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

The frame had a strong effect on efficacy ratings. The frame × age interaction was also significant, but its effect size was very small. None of the demographic variables were significant. Overall, the model had a large effect size.

```r
sum.mod1$coefficients |>
  knitr::kable(digits = 3)
```

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	4.374	0.115	38.178	0.000
frame1neg	-0.648	0.061	-10.649	0.000
age.c	0.005	0.004	1.235	0.217
genderMale	-0.020	0.063	-0.313	0.755
genderOther	0.269	0.240	1.123	0.262
ethnicityBlack or African American	0.181	0.116	1.564	0.118
ethnicityAsian	-0.051	0.119	-0.426	0.670
ethnicityOther	-0.256	0.159	-1.609	0.108

	Estimate	Std. Error	t value	Pr(>|t|)
educationSome college	0.133	0.118	1.136	0.257
education2 year degree	0.185	0.131	1.405	0.161
education4 year degree	0.120	0.108	1.117	0.264
educationGraduate degree	0.015	0.117	0.128	0.898
politicsRepublican	-0.079	0.080	-0.987	0.324
politicsIndependent	0.075	0.074	1.018	0.309
politicsOther	-0.072	0.149	-0.480	0.631
frame1neg:age.c	-0.010	0.005	-2.041	0.042

At the average age, participants who saw the negative frame rated the drug 0.65 points lower than those who saw the positive frame. For each additional year of age, this gap grew by about 0.01 points.

```
emt.mod1 <- emmeans::emtrends(mod1, ~ frame1, var = "age.c")
emmeans::test(emt.mod1)
```

```
 frame1 age.c.trend      SE  df t.ratio p.value
 pos        0.00456 0.00369 576   1.235  0.2173
 neg       -0.00544 0.00329 576  -1.653  0.0988

Results are averaged over the levels of: gender, ethnicity, education, politics
```

```
pairs(emt.mod1)
```

```
 contrast  estimate     SE  df t.ratio p.value
 pos - neg     0.01 0.0049 576   2.041  0.0417

Results are averaged over the levels of: gender, ethnicity, education, politics
```

Under the positive frame, ratings rose slightly with age, while under the negative frame they fell slightly with age. Neither slope was significant on its own, but the difference between them was. So the framing effect grows with age because of small opposing trends in both frames, not because of the negative frame alone.

```
ages <- c(20, 30, 40, 50, 60, 70)

emm.mod1 <- emmeans::emmeans(mod1, ~ frame1 | age.c,
                             at = list(age.c = ages - mean.age))
pairs(emm.mod1)
```

```
age.c = -20.804:
 contrast  estimate     SE  df t.ratio p.value
 pos - neg     0.44 0.1190 576   3.703  0.0002

age.c = -10.804:
 contrast  estimate     SE  df t.ratio p.value
 pos - neg     0.54 0.0808 576   6.688 <0.0001

age.c =  -0.804:
 contrast  estimate     SE  df t.ratio p.value
 pos - neg     0.64 0.0610 576  10.494 <0.0001

age.c =   9.196:
 contrast  estimate     SE  df t.ratio p.value
 pos - neg     0.74 0.0757 576   9.784 <0.0001

age.c =  19.196:
 contrast  estimate     SE  df t.ratio p.value
 pos - neg     0.84 0.1120 576   7.506 <0.0001
```

```
age.c =  29.196:
 contrast  estimate     SE  df t.ratio p.value
 pos - neg     0.94 0.1550 576   6.052 <0.0001

Results are averaged over the levels of: gender, ethnicity, education, politics
```

The framing effect (positive minus negative) increased steadily with age, from 0.44 points at age 20 to 0.94 points at age 70, and it was significant at every age I tested.

```
dat <- dat |>
  mutate(neg1 = as.numeric(frame1 == "neg"))

mod1.jn <- lm(pick1 ~ neg1 * age + gender + ethnicity + education + politics,
              data = dat)

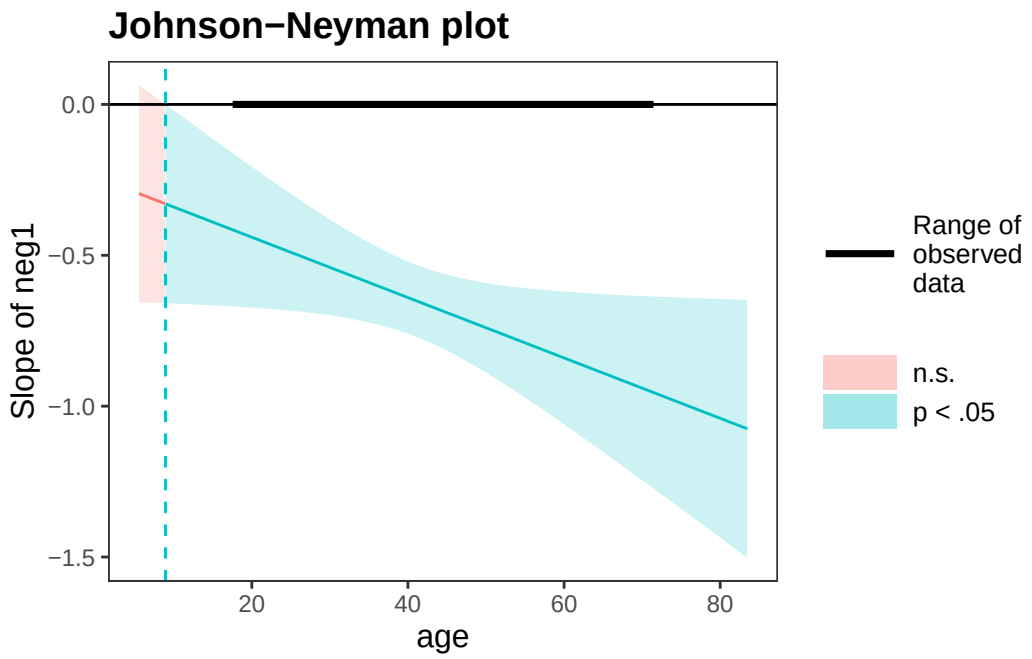
interactions::johnson_neyman(mod1.jn, pred = "neg1", modx = "age",
                             control.fdr = TRUE)
```

```
JOHNSON-NEYMAN INTERVAL

When age is OUTSIDE the interval [-1674.92, 8.94], the slope of neg1 is p <
.05.

Note: The range of observed values of age is [18.00, 71.00]

Interval calculated using false discovery rate adjusted t = 1.96
```

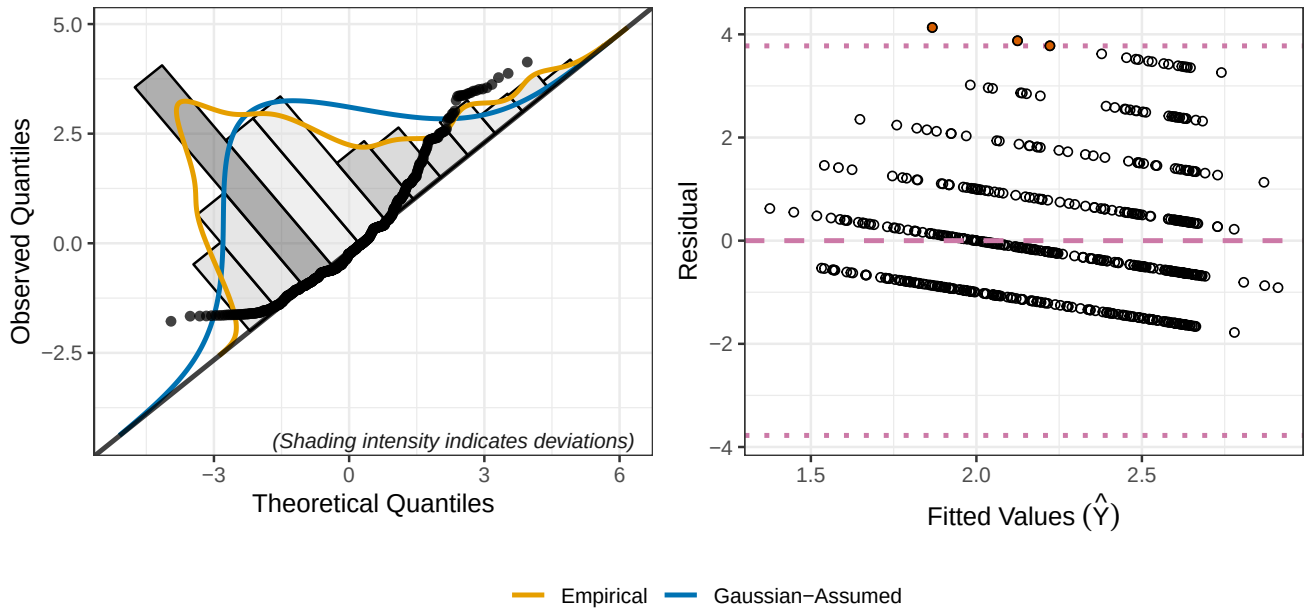


The Johnson-Neyman analysis showed that the framing effect is significant for all ages above 8.94. Since the youngest participant was 18, the framing effect was significant across the entire observed age range.

3.2 Likelihood to Stop Drinking Cocoa

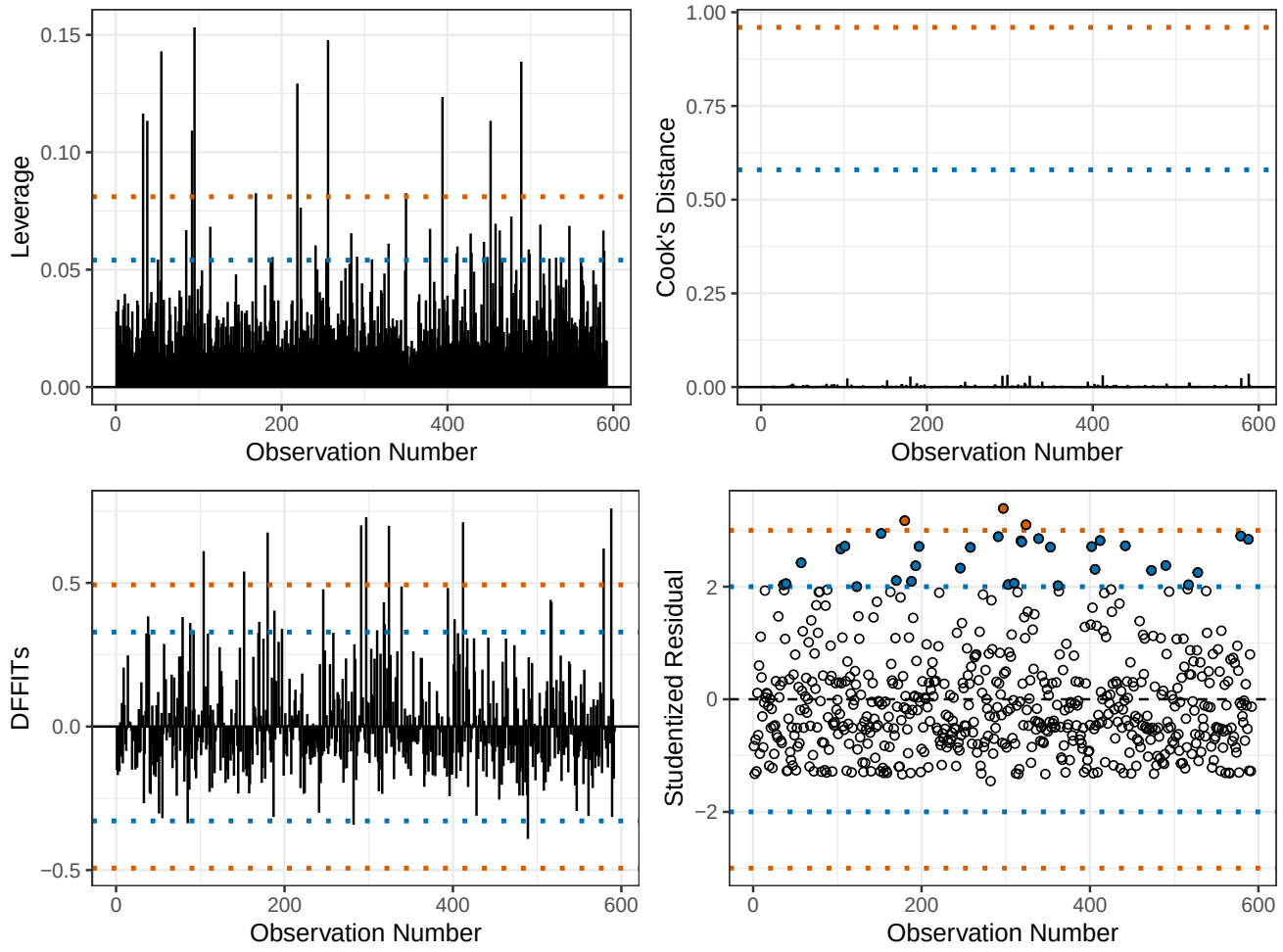
```
mod2 <- lm(pick2 ~ frame2 * age.c + gender + ethnicity + education + politics,
           data = dat)
```

```
plotResiduals(mod2)
```



The residuals again fall along parallel bands because responses are integers from 1 to 6. Residuals are right-skewed: the Q-Q plot shows a heavier upper tail than expected under a Gaussian distribution.

```
plotInfluence(mod2)
```



Some observations have high leverage, but all Cook's distances are far below the moderate threshold, so no observation is highly influential.

```
car::vif(mod2)
```

```
                GVIF Df GVIF^(1/(2*Df))
frame2      1.033200  1        1.016464
age.c       2.168475  1        1.472574
gender      1.076946  2        1.018705
ethnicity   1.145603  3        1.022914
education   1.163222  4        1.019079
politics    1.176610  3        1.027477
frame2:age.c 2.101072  1        1.449508
```

```
car::vif(lm(pick2 ~ frame2 + age.c + gender + ethnicity + education + politics,
            data = dat))
```

```
              GVIF Df GVIF^(1/(2*Df))
frame2    1.033190  1        1.016459
age.c     1.055060  1        1.027161
gender    1.075166  2        1.018284
ethnicity 1.137667  3        1.021729
education 1.148770  4        1.017488
politics  1.176079  3        1.027400
```

All adjusted GVIF values were between 1.0 and 1.5, so multicollinearity is not a concern.

```r
anova(mod2)
```

```
Analysis of Variance Table

Response: pick2
              Df Sum Sq Mean Sq F value    Pr(>F)
frame2         1  48.54  48.545 30.6451 4.712e-08 ***
age.c          1   3.39   3.387  2.1383    0.1442
gender         2   0.82   0.411  0.2592    0.7717
ethnicity      3   2.71   0.902  0.5695    0.6353
education      4   5.74   1.435  0.9056    0.4603
politics       3   0.54   0.181  0.1145    0.9516
frame2:age.c   1   2.23   2.232  1.4092    0.2357
Residuals    576 912.44   1.584
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
sum.mod2 <- summary(mod2)
effectsize::F_to_omega2(f = sum.mod2$fstatistic[1],
                        df = sum.mod2$fstatistic[2],
                        df_error = sum.mod2$fstatistic[3])
```

```
Omega2 (partial) |       95% CI
--------------------------------
0.04             | [0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

```r
effectsize::omega_squared(anova(mod2))
```

```
# Effect Size for ANOVA (Type I)

Parameter    | Omega2 (partial) |       95% CI
-----------------------------------------------
frame2       |             0.05 | [0.02, 1.00]
age.c        |         1.92e-03 | [0.00, 1.00]
gender       |             0.00 | [0.00, 1.00]
ethnicity    |             0.00 | [0.00, 1.00]
education    |             0.00 | [0.00, 1.00]
politics     |             0.00 | [0.00, 1.00]
frame2:age.c |         6.91e-04 | [0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

The frame had a significant effect on willingness to continue drinking cocoa, though the effect size was small. The frame × age interaction was not significant. None of the demographic variables were significant. The overall model had a small effect size ($^2 = 0.04$).

```r
sum.mod2$coefficients |>
  knitr::kable(digits = 3)
```

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	2.357	0.190	12.392	0.000
frame2neg	-0.561	0.105	-5.332	0.000
age.c	-0.001	0.006	-0.207	0.836
genderMale	0.035	0.108	0.321	0.748
genderOther	0.235	0.413	0.570	0.569
ethnicityBlack or African American	-0.233	0.200	-1.169	0.243
ethnicityAsian	-0.082	0.205	-0.400	0.689

	Estimate	Std. Error	t value	Pr(>|t|)
ethnicityOther	-0.052	0.274	-0.190	0.850
educationSome college	0.266	0.201	1.322	0.187
education2 year degree	0.118	0.227	0.517	0.606
education4 year degree	0.275	0.185	1.487	0.138
educationGraduate degree	0.130	0.201	0.646	0.518
politicsRepublican	-0.018	0.138	-0.127	0.899
politicsIndependent	-0.020	0.127	-0.159	0.874
politicsOther	0.127	0.256	0.496	0.620
frame2neg:age.c	-0.010	0.008	-1.187	0.236

At the average age, participants who saw the negative frame were 0.56 points more likely to say they would stop drinking cocoa than those who saw the positive frame. The interaction coefficient is not significant here.

```
emt.mod2 <- emmeans::emtrends(mod2, ~ frame2, var = "age.c")
emmeans::test(emt.mod2)
```

```
 frame2 age.c.trend      SE  df t.ratio p.value
 pos       -0.00127 0.00611 576  -0.207  0.8357
 neg       -0.01124 0.00586 576  -1.918  0.0556

Results are averaged over the levels of: gender, ethnicity, education, politics
```

```
pairs(emt.mod2)
```

```
 contrast  estimate     SE  df t.ratio p.value
 pos - neg  0.00998 0.0084 576   1.187  0.2357

Results are averaged over the levels of: gender, ethnicity, education, politics
```

There is not enough evidence that age changes the framing effect in this scenario.

```
emm.mod2 <- emmeans::emmeans(mod2, ~ frame2 | age.c,
                             at = list(age.c = ages - mean.age))
pairs(emm.mod2)
```

```
age.c = -20.804:
 contrast  estimate    SE  df t.ratio p.value
 pos - neg    0.353 0.204 576   1.734  0.0835

age.c = -10.804:
 contrast  estimate    SE  df t.ratio p.value
 pos - neg    0.453 0.139 576   3.265  0.0012

age.c =  -0.804:
 contrast  estimate    SE  df t.ratio p.value
 pos - neg    0.553 0.105 576   5.246 <0.0001

age.c =   9.196:
 contrast  estimate    SE  df t.ratio p.value
 pos - neg    0.652 0.131 576   4.992 <0.0001

age.c =  19.196:
 contrast  estimate    SE  df t.ratio p.value
 pos - neg    0.752 0.193 576   3.901  0.0001

age.c =  29.196:
 contrast  estimate    SE  df t.ratio p.value
```

```
 pos - neg    0.852 0.267 576   3.188  0.0015

Results are averaged over the levels of: gender, ethnicity, education, politics
```

The estimated framing effect grew from 0.35 points at age 20 to 0.85 points at age 70. It was not significant at age 20 ($p = 0.084$) but was significant from age 30 onward. Because the interaction was not significant, this pattern mostly reflects larger standard errors at the youngest ages rather than a true change in the framing effect.

```
dat <- dat |>
  mutate(neg2 = as.numeric(frame2 == "neg"))

mod2.jn <- lm(pick2 ~ neg2 * age + gender + ethnicity + education + politics,
              data = dat)

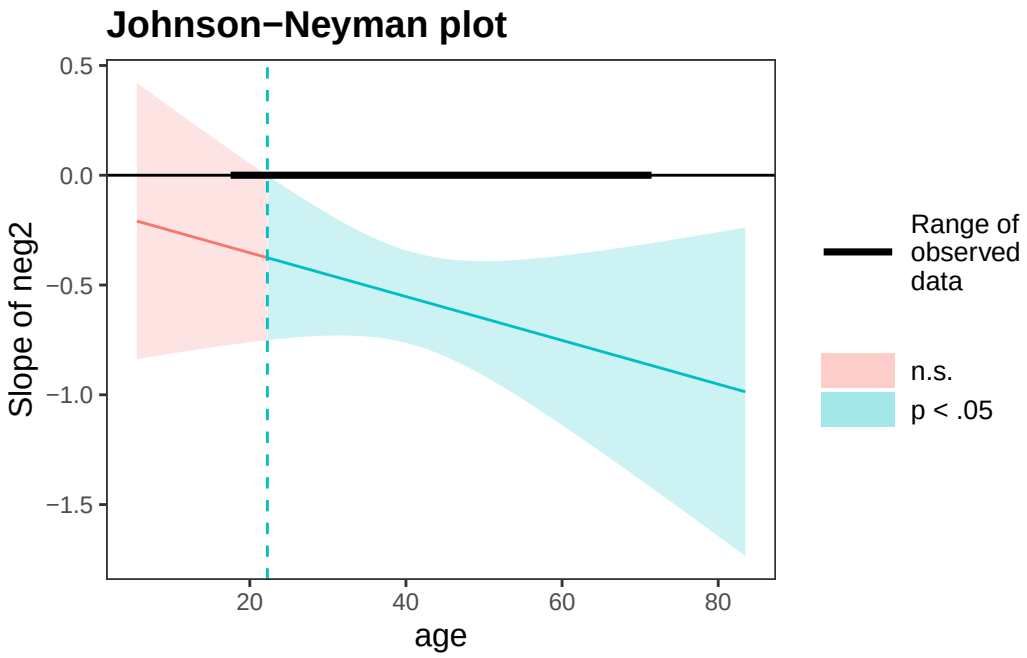
interactions::johnson_neyman(mod2.jn, pred = "neg2", modx = "age",
                             control.fdr = TRUE)
```

```
JOHNSON-NEYMAN INTERVAL

When age is INSIDE the interval [22.25, 120.37], the slope of neg2 is p <
.05.

Note: The range of observed values of age is [18.00, 71.00]

Interval calculated using false discovery rate adjusted t = 2.00
```

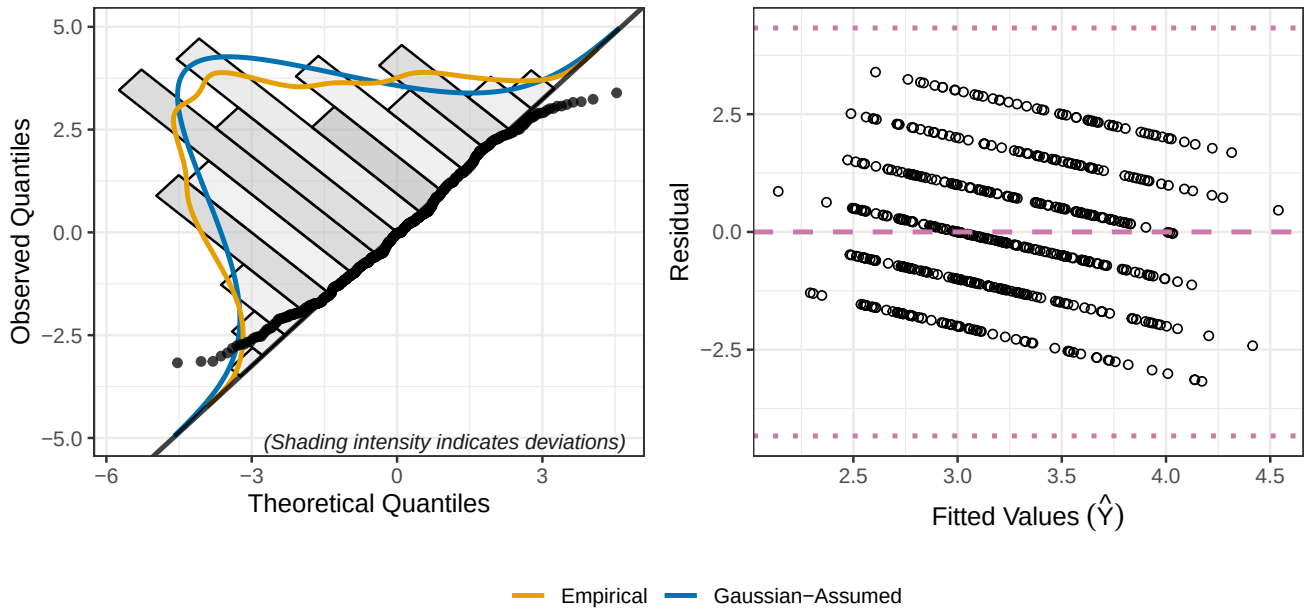


The Johnson-Neyman analysis showed that the framing effect is significant for ages above about 22. However, since the frame × age interaction was not significant, it is not so meaningful to interpret this.

3.3 Preference for Chance Choice

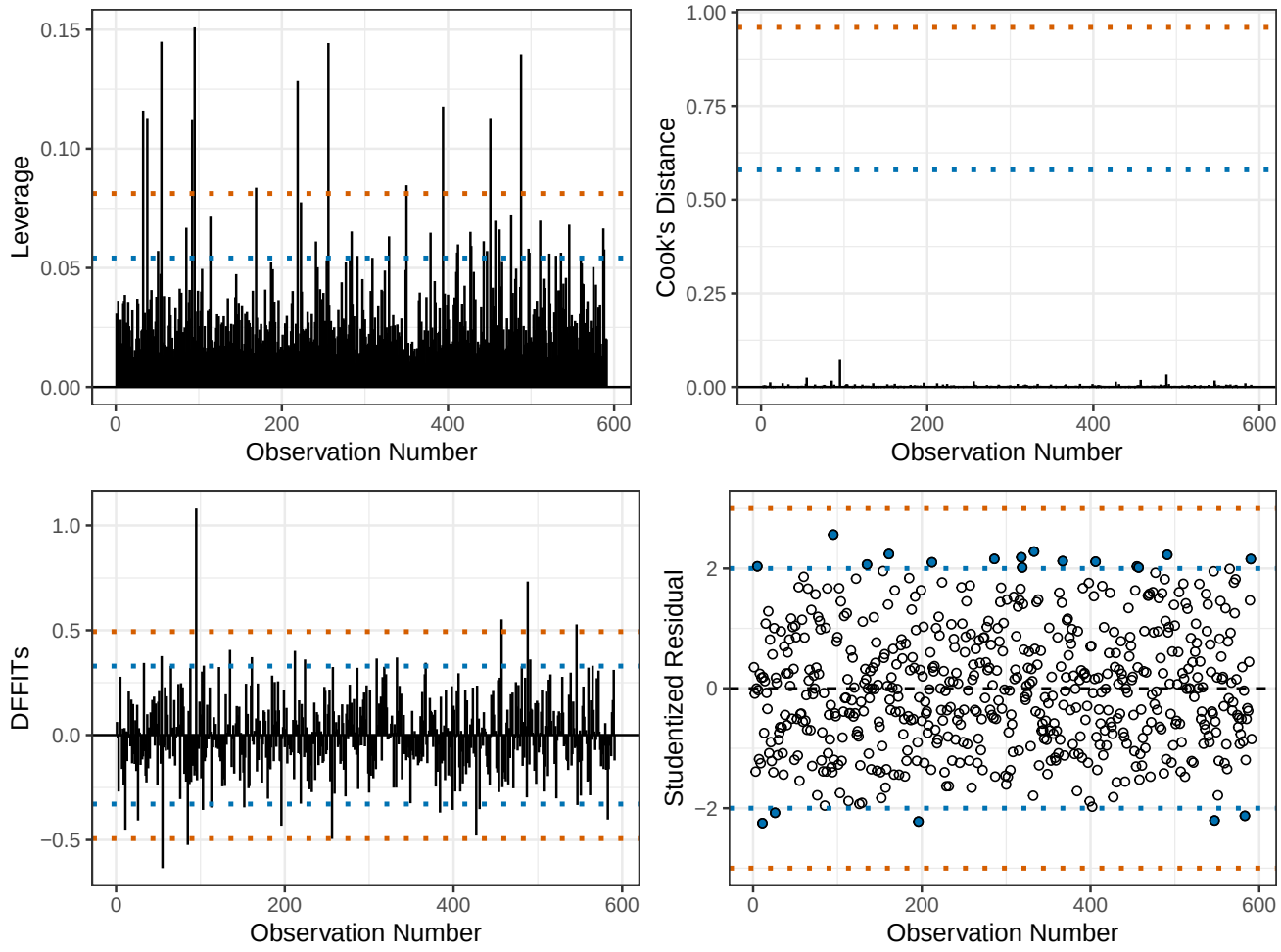
```
mod3 <- lm(pick3 ~ frame3 * age.c + gender + ethnicity + education + politics,
           data = dat)
```

```
plotResiduals(mod3)
```



The residuals fall along parallel bands because responses are integers from 1 to 6. Q-Q plot shows a fair Gaussian distribution.

```
plotInfluence(mod3)
```



Some observations have high leverage, but all Cook's distances are far below the moderate threshold, so no observation is influential.

```
car::vif(mod3)
```

```
                GVIF Df GVIF^(1/(2*Df))
frame3      1.033290  1        1.016509
age.c       2.088709  1        1.445237
gender      1.084603  2        1.020511
ethnicity   1.150019  3        1.023570
education   1.127959  4        1.015165
politics    1.203553  3        1.031361
frame3:age.c 2.027682  1        1.423967
```

```
car::vif(lm(pick3 ~ frame3 + age.c + gender + ethnicity + education + politics,
            data = dat))
```

```
              GVIF Df GVIF^(1/(2*Df))
frame3    1.033230  1        1.016479
age.c     1.061355  1        1.030221
gender    1.076443  2        1.018586
ethnicity 1.136971  3        1.021625
education 1.123259  4        1.014635
politics  1.195032  3        1.030141
```

All adjusted GVIF values were between 1.0 and 1.5, so multicollinearity is not a concern.

```r
anova(mod3)
```

```
Analysis of Variance Table

Response: pick3
              Df  Sum Sq Mean Sq F value   Pr(>F)
frame3         1   66.53  66.528 31.9199 2.53e-08 ***
age.c          1    2.00   2.001  0.9599  0.32762
gender         2    0.78   0.392  0.1882  0.82854
ethnicity      3   18.23   6.075  2.9148  0.03377 *
education      4    3.64   0.909  0.4361  0.78258
politics       3   23.64   7.878  3.7801  0.01050 *
frame3:age.c   1    4.48   4.481  2.1501  0.14311
Residuals    575 1198.42   2.084
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
sum.mod3 <- summary(mod3)
effectsize::F_to_omega2(f = sum.mod3$fstatistic[1],
                        df = sum.mod3$fstatistic[2],
                        df_error = sum.mod3$fstatistic[3])
```

```
Omega2 (partial) |       95% CI
--------------------------------
0.07             | [0.02, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

```r
effectsize::omega_squared(anova(mod3))
```

```
# Effect Size for ANOVA (Type I)

Parameter    | Omega2 (partial) |       95% CI
-----------------------------------------------
frame3       |             0.05 | [0.02, 1.00]
age.c        |             0.00 | [0.00, 1.00]
gender       |             0.00 | [0.00, 1.00]
ethnicity    |         9.63e-03 | [0.00, 1.00]
education    |             0.00 | [0.00, 1.00]
politics     |             0.01 | [0.00, 1.00]
frame3:age.c |         1.94e-03 | [0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

The frame had a significant effect on risk preference, though the effect size was small. The frame × age interaction was not significant. Ethnicity and political affiliation were significant, but their effect sizes were very small. The overall model had a small effect size.

```r
sum.mod3$coefficients |>
  knitr::kable(digits = 3)
```

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	2.487	0.220	11.296	0.000
frame3neg	0.611	0.121	5.061	0.000
age.c	-0.001	0.007	-0.091	0.928
genderMale	0.049	0.124	0.394	0.693
genderOther	0.047	0.475	0.099	0.921
ethnicityBlack or African American	0.361	0.229	1.580	0.115
ethnicityAsian	0.411	0.236	1.737	0.083

	Estimate	Std. Error	t value	Pr(>|t|)
ethnicityOther	-0.417	0.314	-1.329	0.184
educationSome college	0.170	0.231	0.737	0.461
education2 year degree	0.019	0.259	0.073	0.942
education4 year degree	0.244	0.212	1.155	0.248
educationGraduate degree	0.068	0.231	0.292	0.770
politicsRepublican	0.280	0.158	1.768	0.078
politicsIndependent	0.499	0.147	3.386	0.001
politicsOther	0.239	0.295	0.811	0.418
frame3neg:age.c	0.014	0.010	1.466	0.143

At the average age, participants who saw the negative frame were 0.61 points more likely to choose the risky option than those who saw the positive frame. Independents were 0.50 points more likely to choose the risky option than Democrats. The interaction coefficient suggests the framing effect grows with age, but it was not significant.

```
emt.mod3 <- emmeans::emtrends(mod3, ~ frame3, var = "age.c")
emmeans::test(emt.mod3)
```

```
 frame3 age.c.trend      SE  df t.ratio p.value
 pos      -0.000626 0.00689 575  -0.091  0.9276
 neg       0.013565 0.00690 575   1.966  0.0498

Results are averaged over the levels of: gender, ethnicity, education, politics
```

```
pairs(emt.mod3)
```

```
 contrast  estimate      SE  df t.ratio p.value
 pos - neg  -0.0142 0.00968 575  -1.466  0.1431

Results are averaged over the levels of: gender, ethnicity, education, politics
```

Under the positive frame, risk preference did not change with age. Under the negative frame, older participants were slightly more likely to choose the risky option, which is just barely significant. However, the two slopes were not significantly different, so there is only weak evidence that the negative frame drives an age-related change in this scenario.

```
emm.mod3 <- emmeans::emmeans(mod3, ~ frame3 | age.c,
                             at = list(age.c = ages - mean.age))
pairs(emm.mod3)
```

```
age.c = -20.804:
 contrast  estimate    SE  df t.ratio p.value
 pos - neg   -0.316 0.236 575  -1.341  0.1804

age.c = -10.804:
 contrast  estimate    SE  df t.ratio p.value
 pos - neg   -0.458 0.160 575  -2.856  0.0045

age.c =  -0.804:
 contrast  estimate    SE  df t.ratio p.value
 pos - neg   -0.600 0.121 575  -4.954 <0.0001

age.c =   9.196:
 contrast  estimate    SE  df t.ratio p.value
 pos - neg   -0.742 0.149 575  -4.962 <0.0001

age.c =  19.196:
 contrast  estimate    SE  df t.ratio p.value
```

```
 pos - neg   -0.884 0.221 575  -4.002 <0.0001

age.c =  29.196:
 contrast  estimate    SE  df t.ratio p.value
 pos - neg   -1.025 0.306 575  -3.347  0.0009

Results are averaged over the levels of: gender, ethnicity, education, politics
```

The framing effect (negative minus positive) grew from 0.32 points at age 20 to 1.03 points at age 70. It was not significant at age 20 but was significant from age 30 onward.

```r
dat <- dat |>
  mutate(neg3 = as.numeric(frame3 == "neg"))

mod3.jn <- lm(pick3 ~ neg3 * age + gender + ethnicity + education + politics,
              data = dat)

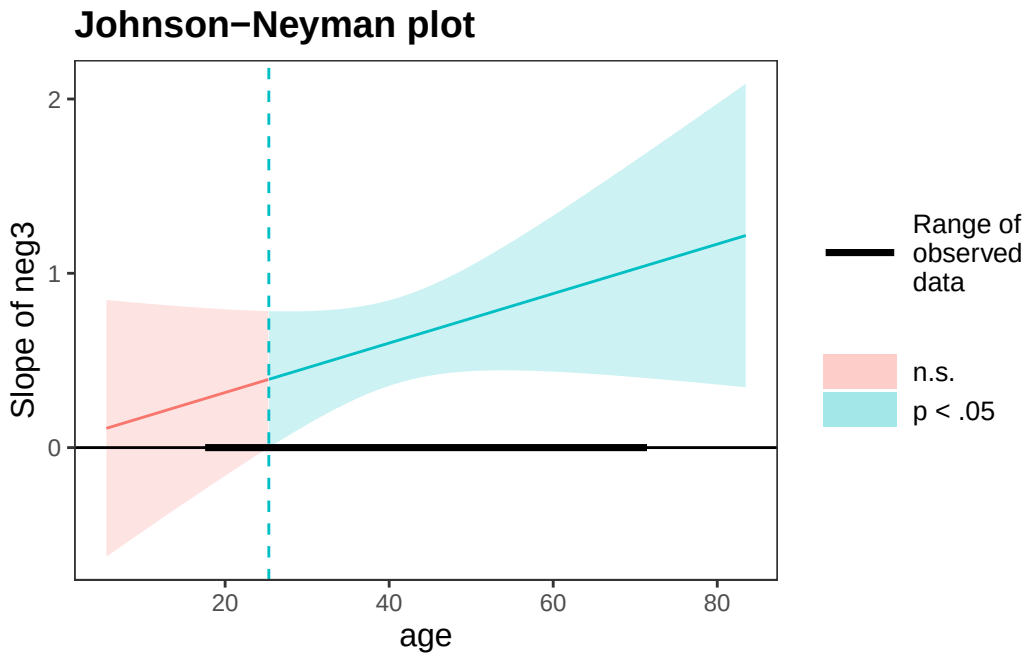
interactions::johnson_neyman(mod3.jn, pred = "neg3", modx = "age",
                             control.fdr = TRUE)
```

```
JOHNSON-NEYMAN INTERVAL

When age is INSIDE the interval [25.33, 151.11], the slope of neg3 is p <
.05.

Note: The range of observed values of age is [18.00, 71.00]

Interval calculated using false discovery rate adjusted t = 2.03
```



The Johnson-Neyman analysis showed that the framing effect is significant for ages between 25.3 and 151.1. This means the effect is significant for participants older than about 25. However, since the frame × age interaction was not significant, I do not interpret 25 as a clear onset age.

4 Discussion

Across all three scenarios, how a choice was worded clearly changed people's responses: the negative frame lowered drug ratings, increased willingness to stop drinking cocoa, and increased preference for the risky option. Whether this framing effect grows with age is less clear. In all three scenarios the interaction pointed in the same direction, with the framing effect larger for older adults, but it was significant only for drug efficacy ($p = 0.042$). In every scenario, the age slope under the positive frame was essentially zero, while the slope under the negative frame was larger and close to significance ($p = 0.099$, 0.056, 0.0498). This pattern is consistent with Xu (2024)'s conclusion that any age-related change is driven by the negative frame, though my evidence for it is weaker.

I did not find a clear onset age. For drug efficacy, the framing effect was significant across the entire observed age range (18–71). For the other two scenarios, the Johnson-Neyman analysis suggested significance after about ag 22–25, but because their interactions were not significant, I do not interpret these as true onset ages. This differs from Xu (2024), who reported onset ages of about 32–34 even though only one of their interactions was significant.

My results may differ from the original study because I analyzed a stricter sample of 592 participants who passed both attention checks, rather than 696.

References

Xu, Lance. 2024. "Advanced Age Predicts Increased Susceptibility to Attribute, Goal, and Risky-Choice Framing in Negative Frame Valences." *Humanities and Social Sciences Communications* 11 (1): 217.